IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

ELECTROMAGNETISM

Wednesday, 6 May 2026, 2:30 PM

Time allowed: 2:00 hours

There are THREE questions on this paper.

Answer ALL questions.

Any special instructions for invigilators and information for candidates are on page 1.

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Formula sheet

Universal constants: $\varepsilon_0 = 8.84 \times 10^{-12}$ F/m $\quad \mu_0 = 4\pi \times 10^{-7}$ A/m $\quad c = 3 \times 10^8$ m/s

Maxwell's equations in integral form

$$\oiint_S \mathbf{D} \cdot \mathrm{d}\mathbf{s} = \iiint_V \rho \mathrm{d}v$$
$$\oiint_S \mathbf{B} \cdot \mathrm{d}\mathbf{s} = 0$$
$$\oint_L \mathbf{H} \cdot \mathrm{d}\mathbf{l} = \iint_S \mathbf{J} \cdot \mathrm{d}\mathbf{s} + \frac{\mathrm{d}}{\mathrm{d}t} \iint_S \mathbf{D} \cdot \mathrm{d}\mathbf{s}$$
$$\oint_L \mathbf{E} \cdot \mathrm{d}\mathbf{l} = -\frac{\mathrm{d}}{\mathrm{d}t} \iint_S \mathbf{B} \cdot \mathrm{d}\mathbf{s}$$

Maxwell's equations in differential form

$$\nabla \cdot \mathbf{D} = \rho$$
$$\nabla \cdot \mathbf{B} = 0$$
$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$
$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

Constitutive relations: $\mathbf{D} = \varepsilon_0 \varepsilon_\mathrm{d} \mathbf{E} \qquad \mathbf{B} = \mu_0 \mu_\mathrm{r} \mathbf{H} \qquad \mathbf{J} = \sigma \mathbf{E}$

Boundary conditions for perfect electric conductors:

$$E_\tau = 0 \qquad E_{n\,\mathrm{out}} = \frac{\sigma}{\varepsilon_0}$$

Boundary conditions for dielectrics:

$$E_{1\tau} = E_{2\tau} \qquad D_{n1} = D_{n2}$$

Biot-Savart law

$$\mathbf{B} = \frac{\mu_0}{4\pi} I \int_L \frac{\mathrm{d}\mathbf{l} \times \mathbf{r}}{r^3}$$

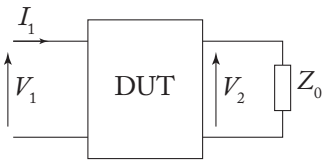
Transmission-line equations

$$\frac{\partial I(z,t)}{\partial z} = -C \frac{\partial V(z,t)}{\partial t} \qquad \text{and} \qquad \frac{\partial V(z,t)}{\partial z} = -L \frac{\partial I(z,t)}{\partial t}$$

Input impedance of a loaded transmission line

$$Z_{\text{in}} = Z_0 \frac{Z_{\text{L}} + jZ_0 \tan(kd)}{Z_0 + jZ_{\text{L}} \tan(kd)}$$

S-parameters for a two-port device



$$S_{11} = \frac{V_1 - Z_0 I_1}{V_1 + Z_0 I_1} \quad \text{and} \quad S_{21} = \frac{2V_2}{V_1 + Z_0 I_1}$$

Fields of a short dipole antenna (far-field region)

$$E_\theta = jZ_0 \frac{I_0 kd}{4\pi} \frac{e^{-jkr}}{r} \sin\theta$$
$$H_\varphi = \frac{E_\theta}{Z_0}$$

Fields of a half-wave dipole antenna (far-field region)

$$E_\theta = \frac{jZ_0 I_0}{2\pi} \frac{\cos\dfrac{\pi\cos\theta}{2}}{\sin\theta} \frac{e^{-jkr}}{r}$$
$$H_\varphi = \frac{E_\theta}{Z_0}$$

Gauss' and Stokes' theorems

$$\oiint_S \mathbf{F} \cdot \mathrm{d}\mathbf{s} = \iiint_V \nabla \cdot \mathbf{F}\, \mathrm{d}v \qquad \oint_L \mathbf{F} \cdot \mathrm{d}\mathbf{l} = \iint_S [\nabla \times \mathbf{F}] \cdot \mathrm{d}\mathbf{s}$$

Useful integrals

$$\int \frac{\mathrm{d}x}{\sqrt{x^2 + a^2}} = \ln(x + \sqrt{a^2 + x^2}) \quad \int \frac{\mathrm{d}x}{(x^2 + a^2)^{3/2}} = \frac{x}{a^2\sqrt{x^2 + a^2}} \quad \int \frac{x\,\mathrm{d}x}{(x^2 + a^2)^{3/2}} = -\frac{1}{\sqrt{x^2 + a^2}}$$

ELECTROMAGNETISM

1. a) The integral form of the continuity equation is

$$\oiint_S \mathbf{J} \cdot \mathrm{d}\mathbf{s} + \frac{dq}{dt} = 0$$

Use this equation to derive a relationship between the voltage and current of a capacitor. [10]

b) The tangential component of the electric field on the surface of a perfect electric conductor (PEC) is zero. Assume static fields. Using the line-integral definition of potential difference, explain why the surface of a PEC must be an equipotential surface. [10]

c) A lossless transmission line has wavelength λ and is terminated in an arbitrary load impedance Z_L. Let $Z_{\text{in}}(d)$ be the input impedance measured a distance d from the load. Show that $Z_{\text{in}}(d + \lambda/2) = Z_{\text{in}}(d)$, and interpret this periodicity in terms of forward and reflected waves. [5]

d) An impedance Z is characterised with a network analyser. Find Z from S_{11}. Assuming Z is a reactance, find $|S_{11}|$ and explain the result. [10]

e) State and briefly explain the difference in the current distributions of a short dipole and half-wave dipole antennas. [5]

2. The vector wave equation for E-field in free space has the form

$$\nabla^2 \mathbf{E} - \mu_0 \varepsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} = 0$$

a) Assume harmonic variation with an angular frequency ω. Also assume propagation in the z-direction only. Show what form the vector wave equation takes. [5]

b) Solve the wave equation derived in part (a). Derive an expression for the wave velocity. Derive a relationship between the wave angular frequency and the wavenumber. [10]

c) Briefly explain what a linearly polarised, transverse electromagnetic wave is. Write down an expression for the E-field of such a wave that propagates in the z-direction. [5]

d) Derive an expression for the H-field of a linearly polarised transverse electromagnetic wave using the expression for the E-field from part (c). [10]

3. a) The fields of a short-dipole antenna in the far-field region have the form

$$E_r = 0, \qquad E_\theta = jZ_0 \frac{I_0 kd}{4\pi} \frac{e^{-jkr}}{r} \sin\theta, \qquad H_\phi = \frac{E_\theta}{Z_0}.$$

Derive an expression for the time-averaged Poynting vector. Which direction does the power flow? Justify the derived distance dependence. [15]

b) The Friis transmission equation is

$$P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t.$$

Define the terms in this equation and briefly explain its physical meaning. [10]

c) Assume G_t and G_r are fixed. If the operating frequency is doubled and the separation R is doubled, by what factor must P_t be changed in order to keep P_r unchanged? [5]